Assignment-4

Aim: To design and study gate-first and gate-last approach for CMOS and effect of variation in different parameters.

Tools used: Semulator 3-D

Theory:
Gate-First Integration:In the traditional gate-first approach, the high-k dielectric and the metal gate are deposited early in the manufacturing process, acting as a mask for the source and drain implants.

- How it works: The actual functional gate stack is formed, followed by the source/drain implantation and a high-temperature activation anneal (typically 900–1050°C).

Gate-Last Integration (Replacement Metal Gate):Also known as the Replacement Metal Gate (RMG) process, gate-last inverts the sequence to protect the sensitive gate materials from extreme heat.

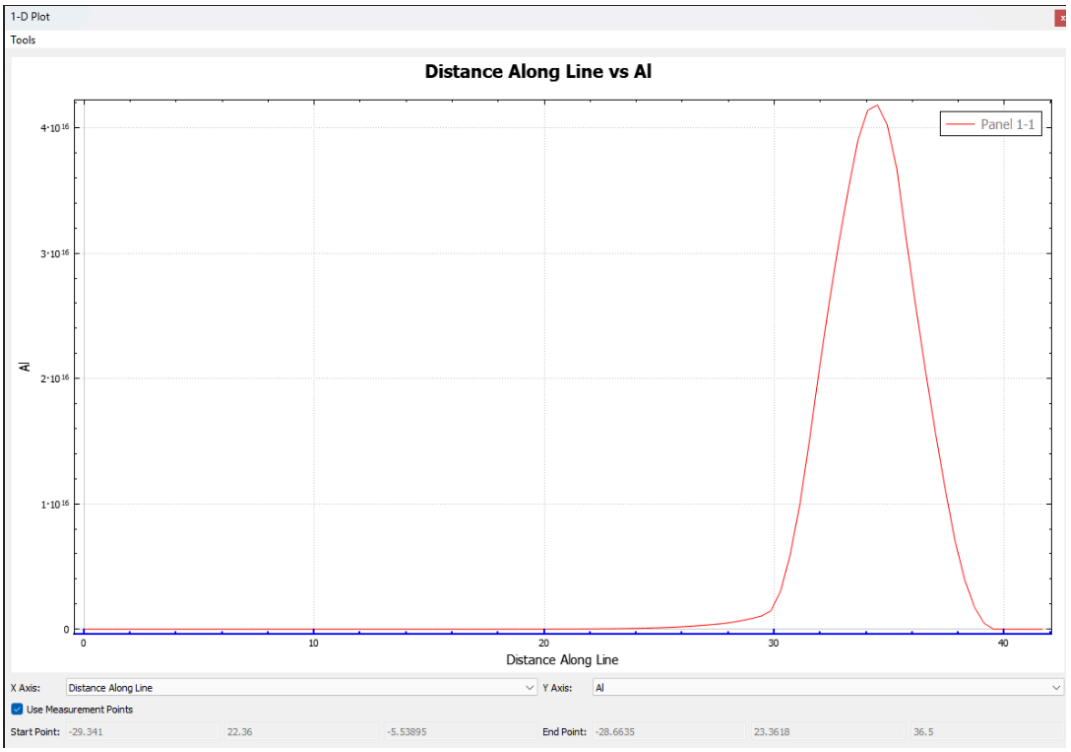
- How it works: A sacrificial "dummy" polysilicon gate is deposited first. The source/drain regions are implanted and annealed at high temperatures using this dummy gate as a mask. Afterward, the dummy gate is selectively etched away to create a trench, and the final high-k dielectric and metal gate materials are deposited into this recess at much lower back-end-of-line (BEOL) temperatures (<400°C). The excess metal is then polished away using Chemical Mechanical Polishing (CMP).

Gate-first approach:

Process steps:

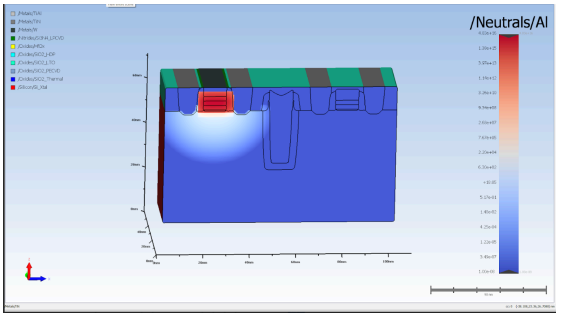
Number	Lock	Action	Name	Material(s)	Dopant(s)	Depth	Mask	Mask Polarity	Wafer	Wafer Side	Lateral Ra
1		Wafer Setup	Wafer Setup	/Silicon/Si_Xtal		50			Wafer1		
2		Conformal Deposit	Pad Oxide Deposition	/Oxides/SiO2_Thermal		5			Wafer1	Top	1.0
3		Conformal Deposit	SiN hardmask	/Nitrides/Si3N4_LPCVD		10			Wafer1	Top	1.0
4		Sequence	STI_Trench_Lithography								
5		Sequence	Etching trench								
6		Remove Materials	resist removal	/Organics/, /Organics/Resist					Wafer1		
7		Interface Growth	Thermal Oxidation of opened trench	/Air, /Silicon/					Wafer1		
8		Conformal Deposit	Filling trench with CVD oxide	/Oxides/SiO2_PECVD		10			Wafer1	Top	1
9		Polish	STI oxide CMP						Wafer1	Top	
10		Enhanced Basic Etch	Removal of SiN hard mask	/Nitrides/Si3N4_LPCVD		10			Wafer1	Top	0
11		Multi Etch (Advanced Modeling)	Pad_oxide ramoval	/Oxides/SiO2_PECVD, /Oxides/SiO2_Thermal		5			Wafer1	Top	
12		Conformal Deposit	Gate oxide deposition	/Oxides/SiO2_HDP		0.5			Wafer1	Top	1.0
13		Conformal Deposit	High k deposition for Gate	/Oxides/HfOx		2			Wafer1	Top	1.0
14		Conformal Deposit	NMOS gate metal	/Metals/TiAl	/Neutrals/Al	2			Wafer1	Top	1.0
15		Sequence	gate patterning								
15.1		Sequence	NMOS Gate_Lithography								
15.2		Enhanced Basic Etch	NMOS Metal Removal to pattern PMOS gate	/Metals/TiAl, /Organics/Resist		2			Wafer1	Top	1
15.3		Remove Materials	resist removal after NMOS gate pattern	/Organics/, /Organics/Resist					Wafer1		
16		Conformal Deposit	gate metal deposition	/Metals/TiN		2			Wafer1	Top	1.0
17		Sequence	NMOS PMOS gates final pattern								
17.1		Conformal Deposit	Final Gate metal deposition	/Metals/W		5			Wafer1	Top	1.0
17.2		Planarizing Deposit	patterning NMOS AND PMOS gates	/Organics/Resist		10			Wafer1	Top	
17.3		Expose Material	Expose for NMOS gate	/Organics/Resist		0	Gate_both_NM...	Light	Wafer1	Top	
17.4		Enhanced Basic Etch	W Etch	/Metals/W, /Organics/Resist		5			Wafer1	Top	0.0
17.5		Enhanced Basic Etch	TiN/TiAl Etch	/Metals/TiAl, /Metals/TiN, /Metals/W, /Organics/Resist, /Oxides/HfOx		5			Wafer1	Top	0.01
17.6		Enhanced Basic Etch	high k Etch	/Metals/TiAl, /Metals/TiN, /Metals/W, /Organics/Resist, /Oxides/HfOx		2			Wafer1	Top	0.0
17.7		Enhanced Basic Etch	Gate oxide Etch	/Metals/TiAl, /Metals/TiN, /Metals/W, /Organics/Resist, /Oxides/HfOx, /Oxides/SiO2_HDP		2			Wafer1	Top	0.0
17.8		Remove Materials	resist removal after gate pattern	/Organics/, /Organics/Resist					Wafer1		
18		Sequence	NMOS source and Drain								
18.1		Conformal Deposit	Conformal Deposit	/Nitrides/Si3N4_LPCVD		3			Wafer1	Top	1.0
18.2		Enhanced Basic Etch	Basic Etch	/Nitrides/		4			Wafer1	Top	0.001
18.3		Planarizing Deposit	Resist pattern for Nmos source and Drain	/Organics/Resist		10			Wafer1	Top	
18.4		Expose Material	Expose resist for source and drain	/Organics/Resist		0	Source_Drain_N...	Dark	Wafer1	Top	
18.5		Implant (Advanced Modeling)	NMOS source and drain doping		/NType/P				Wafer1	Top	
18.6		Remove Materials	resist removal after PMOS gate pattern	/Organics/, /Organics/Resist					Wafer1		
19		Sequence	PMOS source and Drain								
19.1		Planarizing Deposit	Resist pattern for Nmos source and Drain	/Organics/Resist		10			Wafer1	Top	
19.2		Expose Material	Expose resist for source and drain	/Organics/Resist		0	Source_Drain_P...	Dark	Wafer1	Top	
19.3		Implant (Advanced Modeling)	PMOS source and drain doping		/PType/B				Wafer1	Top	
19.4		Remove Materials	resist removal after PMOS gate pattern	/Organics/, /Organics/Resist					Wafer1		
19.5		Diffusion (Advanced Modeling)	Diffusion (Requires Advanced Modeling)	/Oxides/HfOx, /Oxides/SiO2_HDP, /Silicon/Si_Xtal	/NType/P, /Neutrals/Al, /PType/B				Wafer1		
20		Sequence	Source drain metal contacts								

1a. The high-anneal step leads to metal diffusion. Plot the Al metal diffusion in the structures. What is the concentration in the active region?

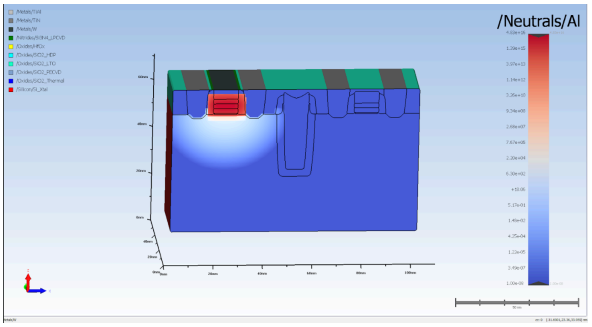


1b. The device stops working if the metal contamination exceeds 1012 cm-3. Go to the anneal recipe and reduce the time of post-implant anneal. What is the maximum time that we can anneal before the contamination exceeds our limit?

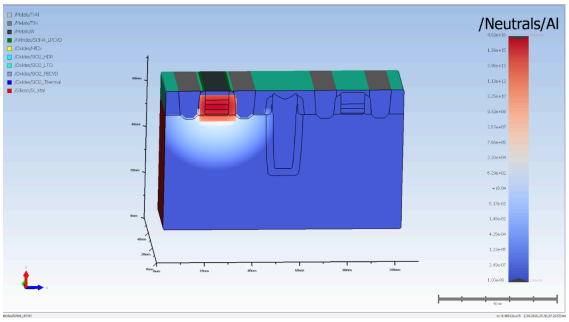
19.5:time	Concentration of Al
6.08	1.35x10^6
6.1	1.43x10^6
6.2	1.65x10^13
6.5	7.07x10^13
7	3.18x10^15
8	3.86x10^15
10	5.57x10^15
12	2.24x10^16



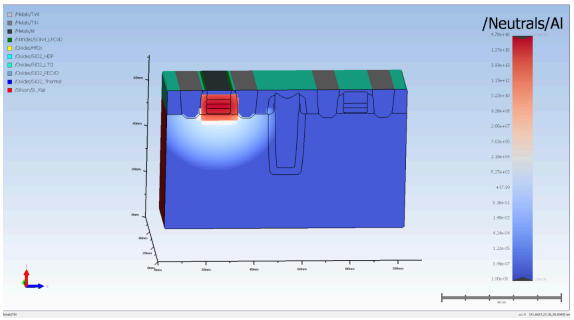
6.1sec



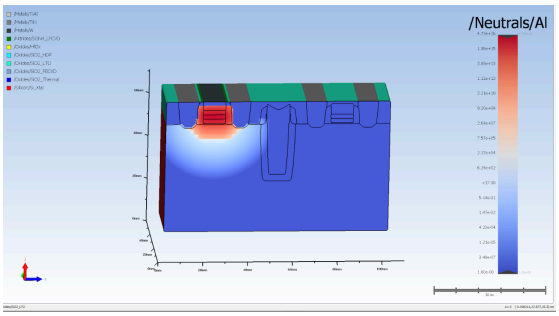
6.08sec



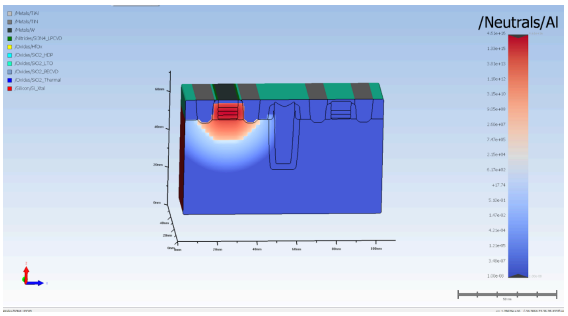
6.2sec



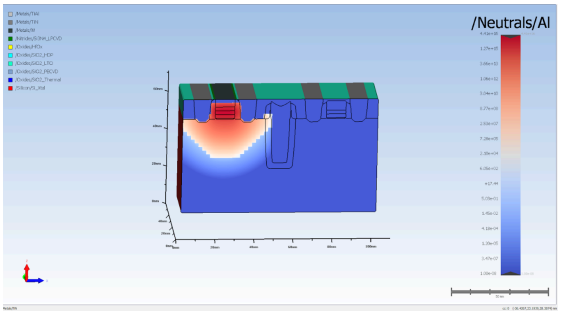
6.5sec



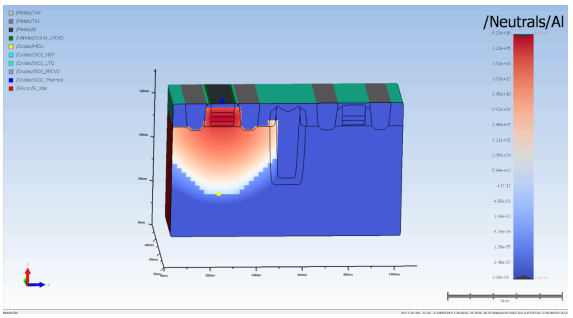
7sec



8sec



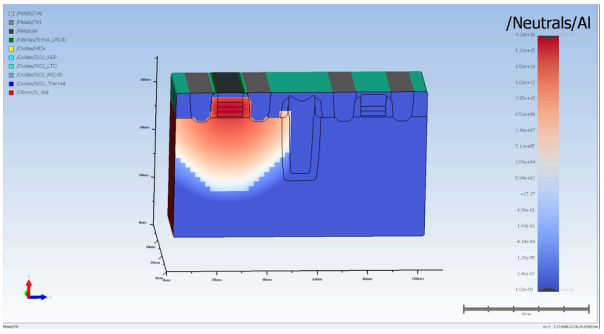
10sec



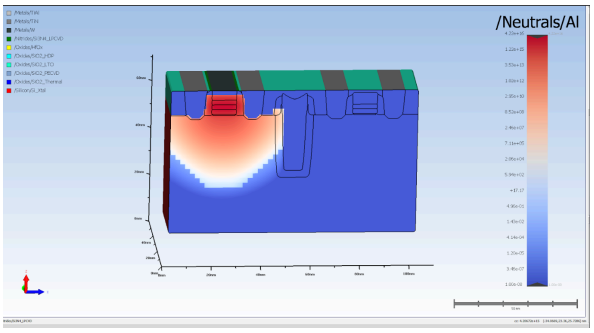
12sec

1c Undo the recipe change of b). Another way to reduce diffusion is to increase gate oxide thickness. What is the minimum thickness of the gate oxide needed to keep metal diffusion below the limit?

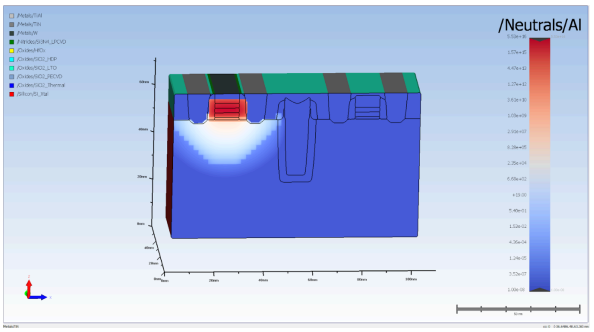
Step-12-Gate oxide deposition	
thickness(in nm)	Concentraion of Al
0.8	1.76x10^15
1	1.2x10^11
2	3.36x10^11
0.4	3.32x10^15
0.5	3x10^15
2.5	9x10^11



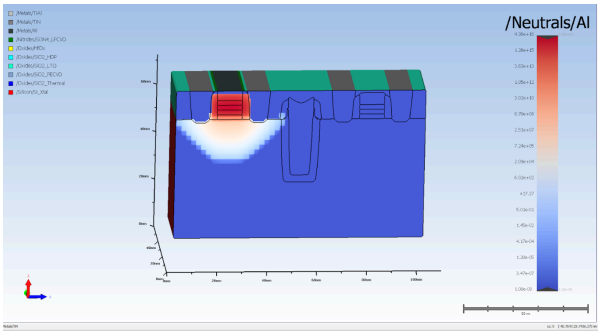
0.4n



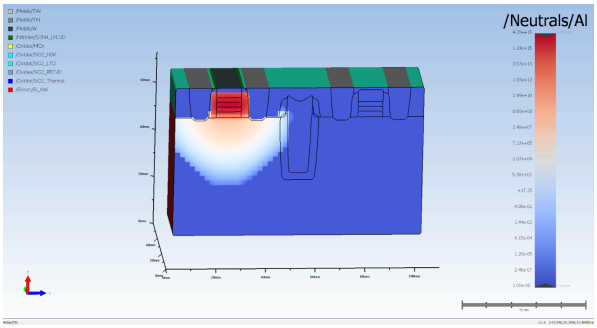
0.5n



1n



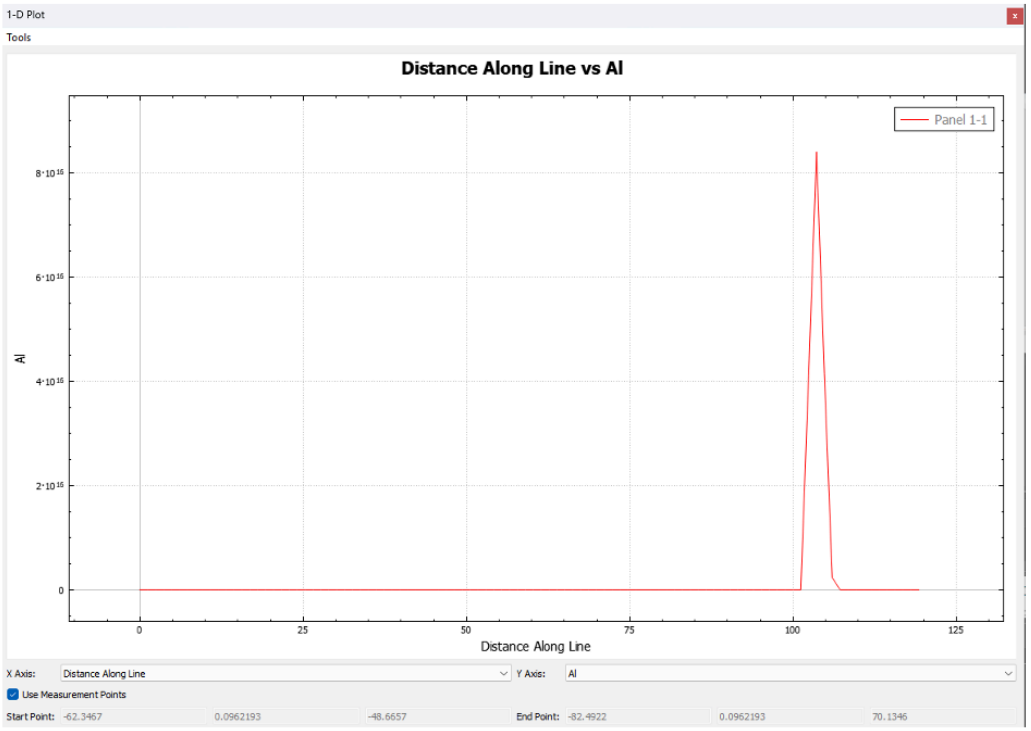
2n



2.5n

Gate-last approach:

2.a The polysilicon dummy gate is refractory, so does not diffuse. Diffusion only occurs during subsequent high-temperature steps. We have added a dummy anneal at the end to represent the combined effect of the future steps. Plot the Al metal diffusion in the structures after the dummy anneal. What is the concentration in the active region?



2b. The device stops working if the metal contamination exceeds 1012 cm-3. Go to the dummy anneal recipe and increase the anneal temperature. What maximum temperature can we anneal before the contamination exceeds our limit?

Anneal Temperature (°C)	Contamination Concentration (cm^{-3})
900	4.00×10^{2} (400)
1000	5.97×10^{5}
1200	2.50×10^{9}
1250	1.32×10^{10}
1275	2.90×10^{10}
1300	6.20×10^{10}

2c. Undo the recipe change of b). The gate last process solves the problem but at the expense of added complexity. List down the various extra steps and materials we Add.

Materials used:

1. Amorphous silicon (Si_Amorph): Used as the sacrificial dummy gate in GateLast.
2. TEOS-based SiO_2: Used as the insulating dielectric surrounding the dummy gate before replacement.

S. No.	**Additional Step/Material**	**Brief Reason**
1	Dummy Gate Deposition (Amorphous Silicon/Polysilicon)	A temporary silicon gate is deposited to maintain the gate structure during high-temperature source/drain processing.
2	Dummy Gate Lithography and Etching	The temporary silicon gate is patterned to define the gate dimensions before it is replaced by metal.
3	Interlayer Dielectric Deposition (SiO_2 using TEOS)	An insulating oxide is deposited around the dummy gates to protect the surrounding structures during gate replacement.
4	Additional Chemical Mechanical Polishing (CMP)	The deposited oxide is planarized to expose the top of the dummy silicon gates for their removal.
5	Poly Pull Etch (Dummy Gate Removal)	The sacrificial silicon gate is selectively removed to create an empty cavity for the permanent metal gate.

6	Replacement Metal Gate Deposition and Patterning	TiAl, TiN and W are deposited after the major thermal processing to form metal gates with the required work functions.
7	Final Metal Gate CMP	Excess metal is removed from the surface, leaving metal only inside the gate cavities and producing a planar surface.

Conclusion: The experiment showed that gate oxide thickness and annealing temperature affect aluminium contamination in the active region. A thinner gate oxide (0.8 nm) resulted in excessive contamination, while thicker oxides provided better protection. Aluminium contamination increased with annealing temperature, but remained within the safe limit up to 1300°C. Thus, choosing a suitable oxide thickness and controlling the annealing temperature are important to minimize aluminium diffusion and ensure reliable device performance.